

| String Length Functions

LENGTH() returns the length of a string.

Depending on the database, this length may represent:

- Number of characters
- Number of bytes

```
SELECT CHAR_LENGTH( 'Aadya' );
```



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Byte Length

Very important.

ASCII

ABC

Each character



1 byte

Result

```
SELECT LENGTH('ABC');
```



3

Unicode



Character count



1

Byte count



4 (UTF-8)

So

```
SELECT LENGTH('😊');
```



4

while

```
SELECT CHAR_LENGTH('😊');
```



1

his is the interview favorite.

Function	Measures
CHAR_LENGTH	Characters
LENGTH	Bytes (MySQL)

Example: 😊😊

Function	Result
CHAR_LENGTH	2
LENGTH	8

Vendor Differences

Database	LENGTH()
MySQL	Bytes
PostgreSQL	Characters
Oracle	Characters
SQLite	Characters

Phone Number

Exactly 10 digits

```
SELECT *  
FROM Customers  
WHERE CHAR_LENGTH(phone)=10;
```

NULL Behavior

```
SELECT LENGTH(NULL);
```



NULL

Regular Expressions (REGEXP)

13.1 Introduction

LIKE is useful for simple patterns.

Examples

Starts with	A%
Ends with	%.com
Contains	%SQL%

But what if you need

- valid email
- only digits
- exactly 10 digits
- starts with capital letter
- optional country code
- hashtags
- URLs

That's where Regular Expressions (Regex) come in.

REGEXP Syntax

MySQL

```
SELECT *  
FROM Users  
WHERE Name REGEXP 'pattern';
```

Example

```
SELECT *  
FROM Users  
WHERE Name REGEXP '^A';
```

Names beginning with A.

Basic Patterns

Any Digit	<code>[0-9]</code>	Any Letter	<code>[A-Za-z]</code>
Uppercase	<code>[A-Z]</code>	Lowercase	<code>[a-z]</code>
Whitespace	<code>[\s]</code>	Digit (MySQL 8+)	<code>[\d]</code>

13.4 Wildcards

`.` means any single character

Example: `A.C`

Matches:

- ABC
- AXC
- A9C

13.5 Quantifiers

Very important.

<code>*</code>	Zero or more	<code>ab*</code>	Matches: a, ab, abb, abbbb
<code>+</code>	One or more	<code>ab+</code>	Matches: ab, abb, abbb
<code>?</code>	Optional	<code>colou?r</code>	Matches: color, colour
<code>{}</code>	Exactly	<code>[0-9]{6}</code>	Exactly six digits.
Range	Between	<code>[0-9]{8,12}</code>	Between 8 and 12 digits.

13.6 Character Classes

<code>[A-Z]</code>	Uppercase	<code>[a-z]</code>	Lowercase
<code>[0-9]</code>	Digits	<code>[A-Za-z0-9]</code>	Alphanumeric

13.7 Anchors

Beginning	<code>^</code>
End	<code>\$</code>

Example
Exactly six digits

```
^[0-9]{6}$
```

13.8 Alternation

Like OR.

```
cat|dog
```

Matches cat or dog

13.9 Grouping

```
(abc)
```

Groups expressions. Useful with alternation.

```
(cat|dog)
```

13.10 Escaping Characters

Need literal	<code>.</code>	Write	<code>\.</code>
Literal	<code>+</code>	↓	<code>\+</code>

REGEXP Functions

REGEXP_REPLACE

Replace using regex.

Example

Remove every digit

```
REGEXP_REPLACE(  
Text, '[0-9]', ''  
)
```

REGEXP_SUBSTR

Extract matching text.

REGEXP_INSTR

Returns the position of the first regex match.

Practical Examples

Validate Email

Very common.

```
WHERE Email REGEXP '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$'
```

Validate Phone

Exactly 10 digits

```
WHERE Phone REGEXP '^[0-9]{10}$'
```

Validate PIN

Exactly six digits

```
WHERE PIN REGEXP '^[0-9]{6}$'
```